Quark Sector Scan: Apr 16

1 Geometry

We use the following metric convention

$$ds^2 = (kz)^{-2}\big(\eta_{\mu\nu}\, dx^\mu\, dx^\nu + dz^2\big), \qquad \eta = \mathrm{diag}(-1, +1, +1, +1), \tag{1}$$

with the UV brane at $z = 1/k$ and the IR brane at $z = 1/\Lambda_{\mathrm{IR}}$.

Two inputs fix everything: the AdS curvature k and the IR scale Λ_{IR}. We set $k = M_{\mathrm{Pl}} = 1.2209 \times 10^{19}$ GeV throughout and scan over Λ_{IR} from 500 GeV to 20 TeV. The warp factor is $\varepsilon = \Lambda_{\mathrm{IR}}/k \sim 10^{-15}$. The Higgs is taken to be localized on the IR brane, and we use $v = 174$ GeV as the electroweak VEV (i.e. $\langle H\rangle = v_{\mathrm{SM}}/\sqrt{2} = 174$ GeV). Backreaction is neglected.

2 Zero-mode profiles

Each bulk fermion has a dimensionless mass parameter c, which controls where its zero mode is localized in the extra dimension. **Possible to do:** We can make this the full integral rather than just the IR overlap The key quantity is the zero-mode's overlap with the IR brane, since the Higgs (and therefore the Yukawa interaction) lives there:

$$f^2_{\mathrm{IR}}(c, \varepsilon) = \frac{\frac{1}{2} - c}{1 - \varepsilon^{1-2c}}\,. \tag{2}$$

This is the point of the RS flavor story: the mass hierarchy comes from the exponential sensitivity of f_{IR} to c, not from hierarchical 5D Yukawa couplings.

3 Yukawa spurion construction

The 5D Yukawa matrices Y_u and Y_d are complex 3×3 matrices. We parametrize each via a singular value decomposition:

$$Y = \texttt{overall\_scale} \times U_{\mathrm{left}} \cdot \mathrm{diag}(\sigma_1, \sigma_2, \sigma_3) \cdot U^\dagger_{\mathrm{right}}, \tag{3}$$

where $\sigma_i > 0$ are ordered light-to-heavy and U_{left} is a unitary parametrized by three angles $(\theta_{12}, \theta_{13}, \theta_{23})$ and a CP phase δ. The prefactor `overall_scale` is a common real positive number multiplying both Y_u and Y_d; it controls the overall magnitude of the Yukawa entries.

We freeze $U_{\mathrm{right}} = I$ for both sectors, which is a restriction: it means $Y^\dagger Y = \mathrm{diag}(\sigma^2)$ is automatically diagonal, while $YY^\dagger = U_{\mathrm{left}}\,\mathrm{diag}(\sigma^2) U^\dagger_{\mathrm{left}}$ is generally dense. Before the fit starts, `overall_scale` is absorbed into the singular values and set to 1.

Setting $U_{\text{right}} = I$ matters because the right-handed bulk mass matrices $C_u = Y_u^\dagger Y_u$ and $C_d = Y_d^\dagger Y_d$ are diagonal by construction, so the dominant source of flavor misalignment is the left-handed matrix C_Q. (Right-handed off-diagonal couplings can still appear after the SVD rotation to the mass basis, but the restriction to diagonal C_u, C_d removes much of the freedom that a general scan would have.) This restricts the scan to 15 free parameters (6 singular values + 8 left-rotation angles + 1 overall scale) out of the 36 that a pair of general 3×3 complex matrices would have.

4 Bulk mass matrices and the role of r

From the Yukawa spurions we build three Hermitian matrices that determine the bulk localization of the quark zero modes:

$$C_u = Y_u^\dagger Y_u, \qquad C_d = Y_d^\dagger Y_d, \qquad C_Q = r\,(Y_u Y_u^\dagger) + (Y_d Y_d^\dagger). \tag{4}$$

In the NMFV literature the full form is $C_Q = \alpha_Q\,\mathbf{I} + \beta_Q\,Y_d Y_d^\dagger + \gamma_Q\,Y_u Y_u^\dagger$, where α_Q is a flavor-universal bulk mass and $r = \gamma_Q/\beta_Q$. This initial scan omits the $\alpha_Q\,\mathbf{I}$ term entirely. **To do**: rerun the scan to include α_Q. This is very simple to do in the code. We can easily change the assumptions and test new models quickly! clamps the eigenvalues into a bounded window after diagonalization. This is not equivalent to including α_Q: it preserves the eigenvectors but distorts the eigenvalue splittings in a way that has no counterpart in the MFV spurion expansion.

The parameter r controls alignment. When $r = 0$, $C_Q \propto Y_d Y_d^\dagger$, which can be diagonalized simultaneously with Y_d. This follows because $U_{\text{right}} = I$, so $Y_d = U_{\text{left}}\,\text{diag}(\sigma)$ and the rotation that diagonalizes $C_Q = Y_d Y_d^\dagger$ also diagonalizes Y_d itself. In that basis the left handed overlap matrix F_Q is diagonal, so there are no tree level FCNCs in the down sector. As r increases, the $Y_u Y_u^\dagger$ term introduces off diagonal entries in C_Q, misaligning the left handed quarks from the down mass eigenbasis and generating tree level FCNCs.

5 From spurion eigenvalues to physical bulk masses

We diagonalize each C_X to get eigenvalues λ_i and a rotation matrix. The question is how to turn those eigenvalues into physical bulk mass parameters c_i. The two frameworks we want to be compatible with handle this differently:

- In 0710.1869, specific benchmark c-values are just quoted directly (Table I), but with "universal terms and overall order one coefficients omitted for simplicity." The bulk masses are input data. This is what I was telling you about earlier.
- In the Csaki et al. (0907.0474) convention, the full $C_Q = \alpha_Q\,\mathbf{I} + \beta_Q\,Y_d Y_d^\dagger + \gamma_Q\,Y_u Y_u^\dagger$ is kept, and the eigenvalues of C_Q are directly the bulk masses. The α_Q term keeps the spectrum in a physical window.

Since this initial scan omits α_Q, the eigenvalues of C_Q are raw non-negative numbers on $[0, \infty)$ with nothing anchoring them in a physical range. To keep all bulk masses inside $[c_{\text{ir}}, c_{\text{uv}}]$ we apply the map

$$c = c_{\text{uv}} - (c_{\text{uv}} - c_{\text{ir}})\,\frac{\lambda}{\lambda + 1}\,, \tag{5}$$

with $c_{\rm uv} = 0.72$, $c_{\rm ir} = 0.30$, giving $c = 0.72 - 0.42\,\lambda/(\lambda+1)$. This maps $\lambda = 0 \to c = 0.72$ (UV-localized), $\lambda \to \infty \to c \to 0.30$ (IR-localized), and is applied identically to all three sectors (Q, u, d).

The choice of $\lambda/(\lambda+1)$ is not arbitrary; Expanding at small λ,

$$c \;\approx\; 0.72 - 0.42\,\lambda \qquad (\lambda \ll 1), \tag{6}$$

which is exactly the linear identification $c = \alpha - \beta\,\lambda$ that the Csaki et al. parametrization would give. Over the actual scan, the C_Q, C_u, C_d eigenvalues have median $\lambda \approx 0.02$, and in that regime the sigmoid and the affine map agree to better than 10^{-3}. The sigmoid only departs from linearity in the large-λ tail (eigenvalues up to ~ 11), where you end up having that the affine map would push $\sim$26% of points below $c_{\rm ir} = 0.30$ into unphysical territory.

The window $[0.30, 0.72]$ is a design choice: $c = 0.30$ gives $f_{\rm IR} \sim 0.45$, which forces the top Yukawa to ~ 2.4 to reproduce $m_t = 172$ GeV (perturbative but above the naive $O(1)$ expectation). Using the same map for all three sectors removes the independent baselines (α_Q, α_u, α_d) that the full parametrization would provide.

Note: With the explicit $(\alpha_Q, \beta_Q, \gamma_Q)$ parametrization the sigmoid goes away entirely. That is, eigenvalues are directly the bulk masses. The sigmoid is an assumption that is likely to bias the top sector masses and the FCNC bounds.

Note: The entire analysis is tree-level. In the Csaki et al. (0907.0474) "shining" framework, up type Yukawa loops induce radiative flavor misalignment, generating off-diagonal down type Yukawas and kinetic mixings that can be numerically important near current bounds.

6 Zero-mode overlaps, basis rotation, and mass matrices

With the physical bulk masses in hand, we compute the IR-brane overlaps $F_Q[i] = f_{\rm IR}(c_Q[i], \varepsilon)$ for the left-handed doublets, and similarly F_u, F_d for the right-handed singlets. We then rotate the Yukawas into the bulk-mass eigenbasis, and the effective 4D quark mass matrices are then

$$M_{u,d} = 2v \cdot \mathrm{diag}(F_Q) \cdot Y^{\rm bulk}_{u,d} \cdot \mathrm{diag}(F_{u,d}), \tag{7}$$

where $2v = 348$ GeV $\approx \sqrt{2}\,v_{\rm SM}$. The 5D Yukawas in this convention are dimensionless.

7 Physical masses, CKM, and the fit

Each mass matrix is decomposed by SVD: $M = U_L\,\mathrm{diag}(m_1, m_2, m_3) U_R^\dagger$, giving the physical quark masses as singular values. The CKM matrix is $V_{\rm CKM} = (U_L^u)^\dagger U_L^d$, from which we extract $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$, and the Jarlskog invariant J.

We wrap the whole chain (Yukawa parameters $\to$ masses + CKM) in an optimizer that adjusts 14 free parameters (6 log-singular-values + 8 left-rotation angles) to match the observed quark masses and CKM elements. The residual is a 10-component vector: 6 log-mass ratios plus 4 fractional CKM deviations. We define the fit score as the RMS of these 10 residuals; points with score > 0.1 (roughly $> 10\%$ average deviation) are discarded.

The target masses are hard coded values chosen for scan stability, not $\overline{\rm MS}$ masses at a consistent scale.

To do: Use PDG $\overline{\rm MS}$ masses at a consistent renormalization scale.

8 The scan

The scan sweeps three parameters on a grid:

Parameter	Grid	Range	Points
r	log-uniform	0.02–2.0	100
`overall_scale`	linear	1.5–6.0	20
Λ_{IR}	log-uniform	500–20 000 GeV	50

r controls the degree of alignment in C_Q (previous sections). `overall_scale` sets the overall Yukawa magnitude. Different values lead to different internal Yukawa structures even after the fitter re-optimizes to hit the same quark masses: if `overall_scale` is large, all Yukawa entries are large and the fitter has to make the singular values small to compensate; if it is small, the singular values have to be large. These different configurations produce different off-diagonal entries in the mass basis and therefore different FCNC predictions, even though the diagonal masses come out the same. Since the fitter freely adjusts all six singular values, `overall_scale` is formally reabsorbable and is not an independent physical degree of freedom. Its role is just that of a scan hyperparameter, since different values seed the optimizer in different basins of the fit landscape.

Over the scan range 1.5 to 6.0, the largest singular value (top-like) runs from ~ 1.5 to ~ 6 and the smallest (up-like) from ~ 0.005 to ~ 0.02. At the high end the top Yukawa is well above the naive $O(1)$ expectation. Λ_{IR} sets the KK scale and therefore the $1/M_{\text{KK}}^2$ suppression of all Wilson coefficients.

Total: $100 \times 20 \times 50 = 100{,}000$ points, each fitted independently from the same deterministic seed.

At scan time several quantities are held fixed. The KK-mass convention is $\xi_{\text{KK}} = 1.0$, meaning $M_{\text{KK}} = \Lambda_{\text{IR}}$ in the Wilson coefficient propagator; the enhanced KK-gluon coupling is $g_s^* = 3.0$.

To do: Future scans should set ξ_{KK} at the physical first KK-gluon mass from the gauge-sector Bessel root) so that $M_{\text{KK}} = m_g^{(1)}$ from the start.

To do: Confirm g_s^* with Lisa. Tree-level $g_s^* \sim 6$ vs. one-loop $g_s^* \sim 3$ is a factor of 4 in ${g_s^*}^2$.

9 KK-gluon couplings in the mass basis

After the fit converges we have the diagonal overlaps F_Q, F_u, F_d and the left/right rotation matrices $U_L^{u,d}$, $U_R^{u,d}$ from the SVD. We model the KK-gluon coupling to zero-mode quarks as proportional to $\text{diag}(F^2)$ in the bulk-mass eigenbasis. In the physical mass basis this picks up off-diagonal entries from the rotations:

$$g_L^{\text{down}}[i,j] = g_s^* \cdot \left(U_L^{d\,\dagger}\,\text{diag}(F_Q^2)\,U_L^d\right)_{ij}, \tag{8}$$

and analogously for g_R^{down} (using U_R^d, F_d), g_L^{up} (using U_L^u, F_Q), and g_R^{up} (using U_R^u, F_u). Note that F_Q enters both LH couplings because u_L and d_L share the same $SU(2)$ doublet.

10 Wilson coefficients

Integrating out the KK-gluon at tree level at the scale $\mu = M_{\text{KK}}$ gives four $\Delta F = 2$ operators. The operators below are labeled $(Q_4^{\text{LR}}, Q_5^{\text{LR}})$, which is the SUSY basis convention (Gabbiani et al.), not the BMU basis ($Q_1^{\mathrm{LR}}, Q_2^{\mathrm{LR}}$). The anomalous dimension matrix in Section 10 and the matrix elements in Section 12 are also written in the SUSY basis.

$$C_1^{\mathrm{VLL}} = \frac{(g_L[i,j])^2}{6\,M_{\mathrm{KK}}^2}\,, \qquad C_1^{\mathrm{VRR}} = \frac{(g_R[i,j])^2}{6\,M_{\mathrm{KK}}^2}\,, \tag{9}$$

$$C_4^{\mathrm{LR}} = -\frac{g_L[i,j]\,g_R[i,j]}{M_{\mathrm{KK}}^2}\,, \qquad C_5^{\mathrm{LR}} = \frac{g_L[i,j]\,g_R[i,j]}{3\,M_{\mathrm{KK}}^2}\,. \tag{10}$$

The five meson systems and their quark-pair indices are: ϵ_K and ΔM_K use (d,s); B_d uses (d,b); B_s uses (s,b); D^0 uses (u,c). We include only the first KK-gluon mode (Possible To do: we can include higher modes for the subleading correction) and use $M_{\mathrm{KK}} = \Lambda_{\mathrm{IR}}$ as both the propagator mass and matching scale.

11 QCD running

The Wilson coefficients are matched at M_{KK} and the hadronic matrix elements are evaluated at $\mu_{\mathrm{had}} = 2$ GeV, so we evolve the coefficients down using leading log QCD RG equations. The LR operators C_4 and C_5 mix via the 2×2 anomalous dimension matrix

$$\gamma_{\mathrm{LR}} = \begin{pmatrix} 8 & -4 \\ -16/3 & -28/3 \end{pmatrix}. \tag{11}$$

This is the anomalous dimension matrix in the SUSY (C_4, C_5) basis, not the BMU $(C_1^{\mathrm{LR}}, C_2^{\mathrm{LR}})$ basis, which matches the code. The dominant eigenvalue is large and positive, giving a 3 to 5$\times$ enhancement of the LR operators at low scales. Combined with the chiral enhancement factor $r_\chi = (m_K/(m_s+m_d))^2 \approx 25$ in the kaon matrix elements, this is why ϵ_K is the most powerful constraint.

The running crosses one flavor threshold at $m_b = 4.18$ GeV ($n_f = 5 \to 4$); $m_c = 1.27$ GeV is below μ_{had}. We use one-loop α_s running with $\alpha_s(M_Z) = 0.1179$ and continuous threshold matching.

12 Hadronic matrix elements and observables

The evolved Wilson coefficients at $\mu_{\mathrm{had}} = 2$ GeV are contracted with hadronic matrix elements to get the NP mixing amplitude:

$$M_{12}^{\mathrm{NP}} = C_1^{\mathrm{VLL}} \cdot \mathrm{ME}_{\mathrm{VLL}} + C_1^{\mathrm{VRR}} \cdot \mathrm{ME}_{\mathrm{VLL}} + C_4^{\mathrm{LR}} \cdot \mathrm{ME}_{\mathrm{LR4}} + C_5^{\mathrm{LR}} \cdot \mathrm{ME}_{\mathrm{LR5}}, \tag{12}$$

where VRR and VLL share the same matrix element by parity.

To do: I am pretty sure I have a factor of 2 subtlety here I need to fix. The matrix elements are

$$\mathrm{ME}_{\mathrm{VLL}} = \tfrac{2}{3} f_P^2 m_P B_1, \qquad \mathrm{ME}_{\mathrm{LR4}} = \left(\tfrac{r_\chi}{6} + \tfrac{1}{4}\right) f_P^2 m_P B_4, \qquad \mathrm{ME}_{\mathrm{LR5}} = \left(\tfrac{r_\chi}{2} + \tfrac{1}{12}\right) f_P^2 m_P B_5, \tag{13}$$

with $r_\chi = (m_P/(m_{q_1}+m_{q_2}))^2$. The matrix elements above already absorb the $1/(2m_P)$ state-normalization factor, so $M_{12} = \sum_i C_i \cdot \mathrm{ME}_i$ is the standard off-diagonal mass-matrix element in GeV and $\Delta m = 2|M_{12}|$. The code computes $\mathrm{ME}_{\mathrm{VLL}} = \frac{2}{3}\, f_P^2\, m_P\, B_1$ and contracts it directly with C_i to obtain M_{12}, but elsewhere we compute $\langle Q_1 \rangle = \frac{8}{3}\, f_P^2\, m_P^2\, B_K$ and divides by $2m_P$ explicitly. With the same $B_1 = B_K$.

Hadronic inputs (decay constants, bag parameters, quark masses) come from FLAG 2024 and PDG 2024, with the exception of the D^0 LR bag parameters B_4, B_5 which are estimated at 1.0.

13 Acceptance criteria

For each parameter point we compute the KK-gluon contribution to five meson mixing observables: ϵ_K, ΔM_K, ΔM_{B_d}, ΔM_{B_s}, ΔM_{D^0}. Each one gets a "budget," the amount of room the experimental measurement leaves for new physics beyond the SM. We form the ratio

$$\text{ratio} = \frac{\text{KK-gluon prediction}}{\text{budget}}\,, \tag{14}$$

and a point passes only if all five ratios are ≤ 1 The budgets are deliberately generous, so these bounds should be read as a weak "best-case envelope," not a sharp exclusion.

For ϵ_K, the observable is $\epsilon_K^{\text{NP}} = |\kappa_\epsilon/(\sqrt{2}\,\Delta m_K)| \cdot |\text{Im}(M_{12}^{\text{NP}})|$ with $\kappa_\epsilon = 0.94$, and the budget is the exp–SM gap: $|2.228 \times 10^{-3} - 1.81 \times 10^{-3}| = 4.18 \times 10^{-4}$. The absolute value discards the sign of the so there is no constructive/destructive interference with SM.

For ΔM_K, the budget is $\Delta m_K^{\text{exp}}/2$ (the SM prediction is long distance dominated and unreliable, so the entire experimental value is treated as room for new physics).

For B_d and B_s, the budget is $\max(\Delta m^{\text{exp}}/2,\ |\Delta m^{\text{exp}} - \Delta m^{\text{SM}}|/2)$, which in practice is dominated by $\Delta m^{\text{exp}}/2$. For D^0 it is simply $\Delta m^{\text{exp}}/2$.

To do: I think there is a lot to improve here. Right now we only check whether the RS contribution fits inside this defined budget for each system independently.

14 Post-processing and figures

Because the scan used $M_{\text{KK}} = \Lambda_{\text{IR}}$ as the propagator mass, but the physical first KK-gluon mass is $m_g^{(1)} = \xi_{\text{kk}}\Lambda_{\text{IR}}$ with $\xi_{\text{kk}} = 2.4487$ (the first gauge-sector Bessel root), the plotting script relabels the axis to $m_g^{(1)}$ and divides all ratios by ξ_{kk}^2 to correct the propagator. This does not correct the QCD running start scale (see the TODO above).

At each (r, M_{KK}) grid point there are up to 20 `overall_scale` values. The plotter picks the single `overall_scale` that minimizes max(all 5 ratios), so that all five contour lines in the figure come from one physically consistent MFV point.

Both figures are best-case envelopes: the fitter picks the most favorable Yukawa structure at each grid point, and the plotter picks the most favorable `overall_scale`. So the contour shows the lowest M_{KK} at which any fitted point survives; it is not a statement about typical NMFV behavior.

To do: These figures answer "can NMFV ever allow M_{KK} this low?" Your note asks for an ensemble analysis: random $O(1)$ Yukawas in the explicit $(\alpha_Q, \beta_Q, \gamma_Q)$ parametrization, keep points reproducing quark masses, histogram M_{KK} where $\text{Im}(C_4^K)$ meets its bound. Both analyses are needed.